

AWS DevOps Master Program

Job-Oriented Training | Build. Automate. Deploy. Master the full DevOps pipeline from Git to Kubernetes — and land your dream job in tech.

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Introduction to DevOps

What is DevOps?

DevOps is a cultural and technical movement that unites software development (Dev) and IT operations (Ops) to deliver software faster, more reliably, and at scale. It bridges the gap between writing code and running it in production.

Rooted in Agile principles, DevOps emerged to eliminate bottlenecks in the traditional waterfall model — enabling teams to ship features continuously without sacrificing stability.

Core Objectives

→ Accelerate Delivery

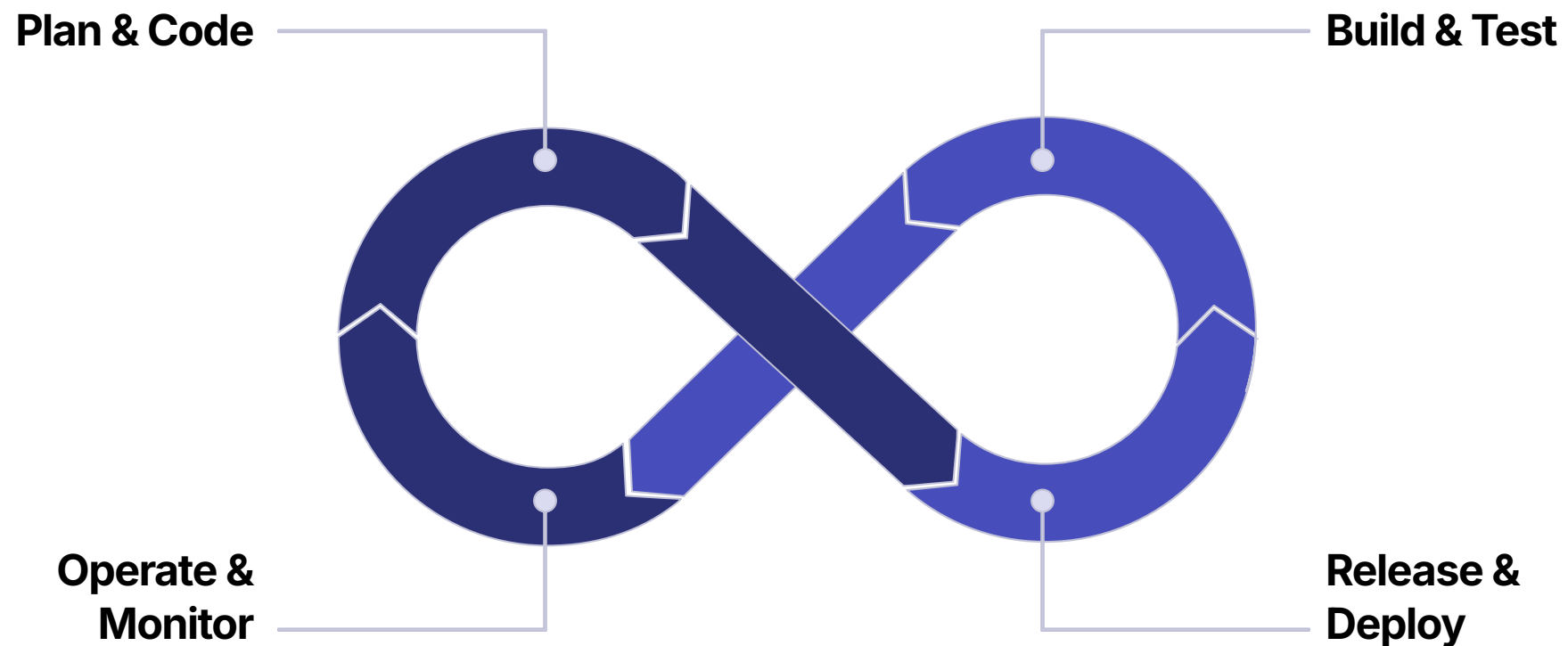
Automate build, test, and release pipelines for faster time-to-market.

→ Improve Collaboration

Break silos between Dev, QA, and Ops teams with shared ownership.

→ Ensure Stability

Implement continuous monitoring and feedback loops for reliable systems.



The DevOps lifecycle creates a continuous feedback loop — every phase feeds into the next, enabling teams to iterate rapidly and respond to change with confidence.

Git — Source Code Management

Git is the industry-standard version control system. Master everything from basic commits to advanced branching strategies used by top engineering teams worldwide.

Core Concepts

- Git vs GitHub vs SVN
- Git stages & lifecycle
- Repository setup & configuration
- Push, pull, fetch & stash

Branching & Merging

- Git branches & strategies
- Git merge vs Git rebase
- Resolving Git conflicts
- Cherry-pick & revert

Collaboration

- Pull requests & code reviews
- Git tags & versioning
- Git webhooks
- Protecting GitHub branches

Maven & Tomcat — Build & Deploy

Maven — Build Automation

Maven is the backbone of Java project builds. Learn to manage your entire project lifecycle — from compiling code to packaging deployable artifacts and managing third-party dependencies through centralized repositories.

Lifecycle & Phases

Understand compile, test, package, install, and deploy phases.

Artifacts

Generate .jar, .war, and .ear files for different deployment targets.

Dependencies

Add plugins, properties, and libraries via pom.xml configuration.

Tomcat — Application Server

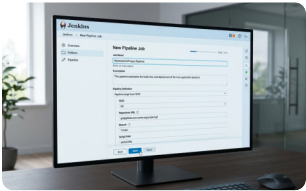
Apache Tomcat is the go-to application server for deploying Java web applications. You'll install and configure Tomcat on both Windows and Linux/AWS environments, then learn to deploy code manually and via Jenkins automation.

- Install Tomcat on Windows & Linux/AWS
- Deploy .war files to Tomcat server
- Automate deployments through Jenkins
- Integrate with CI/CD pipelines end-to-end

MODULE 5

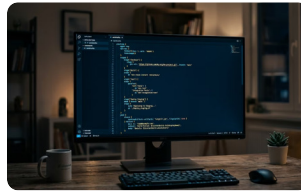
Jenkins — CI/CD Automation Engine

Jenkins is the heart of any DevOps pipeline. From creating your first job to building enterprise-grade declarative pipelines, this module gives you complete, job-ready Jenkins mastery.



Pipeline Setup

- Jenkins installation & configuration
- Freestyle, Maven, and Pipeline job types
- Upstream & downstream job chaining
- Parameterized builds & scheduling



Declarative Pipelines

- Jenkinsfile deep dive
- Pipeline syntax & stages
- Shared libraries & plugins
- Blue Ocean plugin & pipeline views



Advanced Administration

- Master-slave node configuration
- Authentication, authorization & security
- SonarQube & Nexus integration
- Jenkins backup & disaster recovery

SonarQube — Code Quality & Security



Deliver Clean, Secure Code Every Time

SonarQube is the industry-standard platform for continuous code quality inspection. It automatically detects bugs, vulnerabilities, code smells, and security hotspots — giving developers instant feedback before code reaches production.

In this module, you'll install and configure SonarQube, connect it to your Maven builds, and integrate it seamlessly into your Jenkins CI/CD pipeline for automated quality gates.

Installation & Setup

Deploy and access the SonarQube server on Linux/AWS environments.

Build Tool Integration

Connect SonarQube with Maven and Jenkins for automated scans.

Quality Gates

Define pass/fail thresholds that block broken code from being merged.

Ansible — Infrastructure Automation

Ansible makes IT automation simple, powerful, and agent-less. Configure hundreds of servers simultaneously using human-readable YAML playbooks — no complex scripting required.



Architecture

Understand Ansible's agentless push model — control node, managed nodes, and inventory structure.



Ad-Hoc Commands

Execute one-time tasks across multiple servers instantly without writing a full playbook.



Playbooks

Write reusable, idempotent YAML playbooks to configure entire environments consistently.



Modules & Roles

Leverage hundreds of built-in modules and organize complex automation into reusable roles.



Docker — Containerization Mastery

Why Docker?

Containers have revolutionized how software is built and shipped. Docker packages your application and all its dependencies into a single portable unit — eliminating the classic "it works on my machine" problem forever.

From writing your first Dockerfile to building production-grade multi-stage images, this is one of the most hands-on modules in the program.

What You'll Master

1 Docker Architecture

Deep dive into Docker engine, daemon, images, containers, registries, and networking.

2 Dockerfile & Image Building

Create custom images, multistage builds, and push to Docker Hub.

3 Docker Compose & Volumes

Orchestrate multi-container apps including a live WordPress implementation.

4 Security with Trivy

Scan Docker images for vulnerabilities before they reach production.

Kubernetes — Container Orchestration at Scale

Kubernetes (K8s) is the undisputed king of container orchestration. Deploy, scale, and manage containerized applications across clusters with enterprise-grade reliability.

Core Objects

- Pods, ReplicaSets & Deployments
- Services & Namespaces
- ConfigMaps vs Secrets
- Volumes, PV & PVC

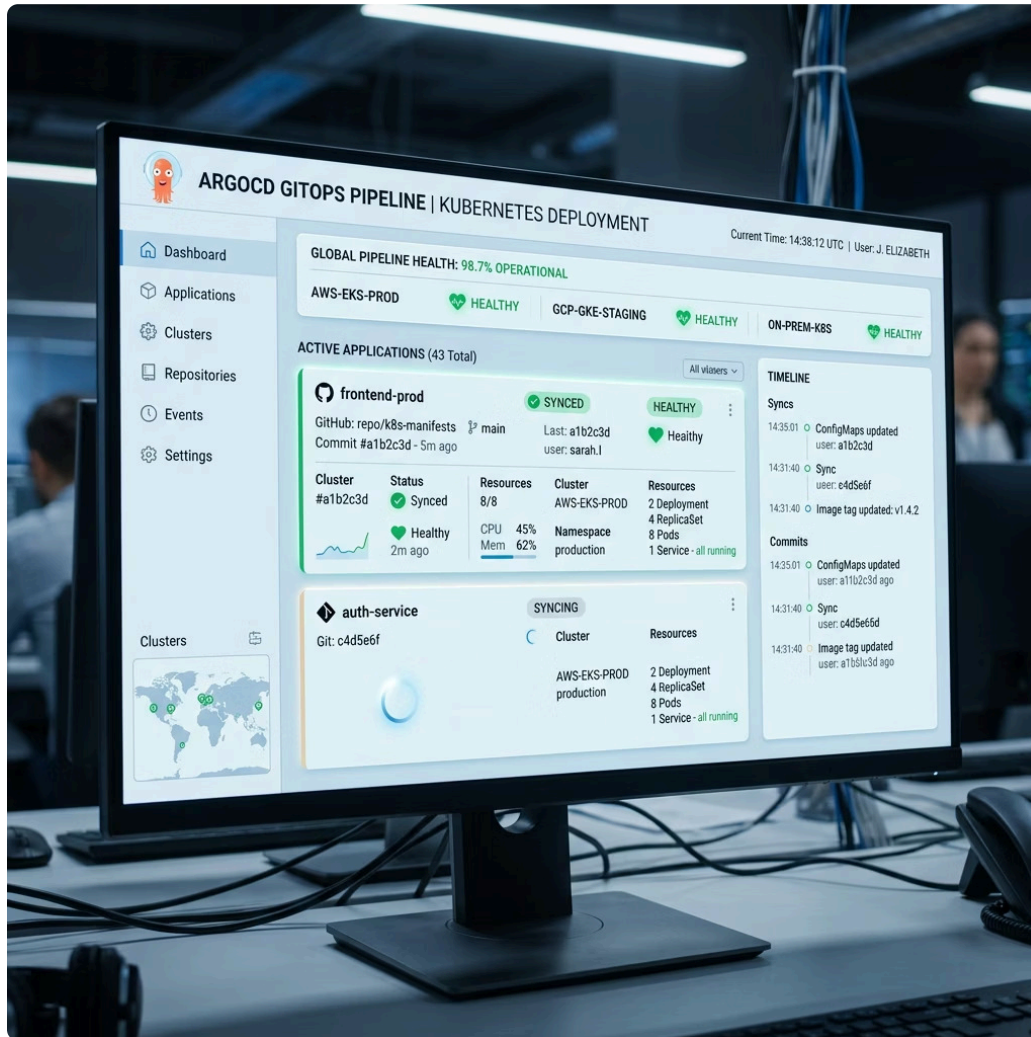
Advanced Features

- RBAC & Security Contexts
- Ingress resources & controllers
- DaemonSets & StatefulSets
- Taints, tolerations & scheduling

AWS EKS & Helm

- EKS cluster creation & upgrades
- Helm charts for deployments
- Network policies
- K8s mini & microservices projects

GitOps with ArgoCD



The Future of Continuous Delivery

GitOps takes CI/CD to the next level by using Git as the single source of truth for infrastructure and application state. Argo CD watches your Git repository and automatically syncs your Kubernetes cluster to match — every change is auditable, reversible, and automated.

01

Install & Configure Argo CD

Deploy Argo CD on a Kubernetes cluster and access the UI.

02

CD Mini Project

Build a real continuous delivery workflow with ArgoCD and K8s.

03

Ultimate CI/CD Pipeline

Combine Git, Jenkins, Maven, Sonar, Docker, EKS, and ArgoCD into one end-to-end production pipeline.

Terraform — Infrastructure as Code

Stop clicking through cloud consoles. With Terraform, you define your entire AWS infrastructure in code — making it repeatable, version-controlled, and team-friendly. This module covers everything from basic resources to production-grade state management.

1

Core Concepts

Architecture, providers, resources, variables, and the Terraform workflow (init → plan → apply → destroy).

2

Modules & Workspaces

Organize reusable infrastructure code with modules and manage multiple environments with workspaces.

3

State Management

Manage remote state files, locking, and backend configuration for team collaboration.

4

Terraform Project

Provision a complete AWS environment — VPC, EC2, RDS, and more — entirely through Terraform code.

Linux — The Foundation of DevOps

Why Linux Matters

Over 90% of cloud servers run Linux. A DevOps engineer who can't navigate the command line is like a pilot who can't read instruments. This module builds a rock-solid Linux foundation that underpins every other tool in the program.

From file permissions to shell scripting, you'll work in real Linux environments from day one.

Skills You'll Build

File & User Management

Create files, directories, users, and groups. Manage permissions with `chmod` and `chown`.

Networking & Remote Access

SSH, SCP, WinSCP, passwordless authentication, and working with services.

Essential Commands

`cp`, `mv`, `tar`, `yum`, `rpm`, `wget`, `find`, `vim`, `cat`, `crontab`, and much more.

Shell Scripting

Write automation scripts to simplify repetitive DevOps tasks and workflows.

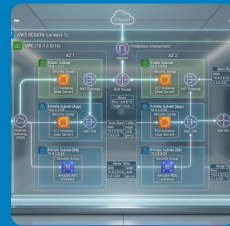
Amazon Web Services (AWS)

AWS is the world's leading cloud platform. This comprehensive module covers the services every DevOps engineer must know — from compute and networking to security and storage.



Compute & Scaling

- EC2 instances, types & AMIs
- Elastic Load Balancers
- Auto Scaling groups
- CloudWatch & billing alarms



Networking & VPC

- Custom VPCs & subnets
- Route tables, IGW & NAT Gateway
- Security Groups vs NACL
- Route 53 & CloudFront



Security & IAM

- IAM users, groups & roles
- Password policies & MFA
- AWS Secrets Manager
- Customer-managed policies



Storage & Databases

- S3 deep dive & EFS
- Volumes, snapshots & EBS
- RDS (MySQL, MSSQL, Aurora)
- SNS & AWS Certificate Manager

Scripting — Shell & Python

Shell Scripting

Shell scripting is the glue that holds DevOps automation together. Learn to write powerful Bash scripts that automate repetitive tasks, manage servers, and integrate with CI/CD pipelines.

→ Basics & Execution

Script syntax, permissions, and running your first shell script.

→ Variables & Input

Dynamic scripts using user input, environment variables, and conditionals.

→ Real-Time Scenarios

Practical automation scripts used in real DevOps workflows.

Python for DevOps

Python is rapidly becoming the most in-demand scripting language for DevOps engineers. Use Python to automate AWS operations, write custom tooling, and integrate with APIs and cloud services.

→ Python Foundations

Syntax, variables, data types, and Python environment setup.

→ Control Structures

Conditionals, loops, functions, and modular programming.

→ Working with Data

File handling, JSON parsing, and working with APIs and modules.

AWS & DevOps Real-World Projects

Theory without practice is incomplete. Every concept you learn is reinforced through hands-on projects that mirror real industry scenarios — the same challenges you'll face on the job from day one.

VPC Mini Project

Design and deploy a secure, custom AWS VPC from scratch with subnets, routing, and gateways.

AWS 2-Tier Architecture

Build a production-style 2-tier application with web and database layers on AWS.

End-to-End 3-Tier Project

Full AWS 3-tier architecture integrating multiple AWS services in a single production-ready deployment.

Ultimate CI/CD Pipeline

Git → Jenkins → Maven → SonarQube → Docker → EKS → ArgoCD — a complete enterprise-grade pipeline.

8+

Hands-On Projects

Real-world AWS and DevOps projects from VPC to microservices

15+

Tools Mastered

From Git to Kubernetes, Terraform, and beyond

100%

Job Focused

Every project mirrors real industry scenarios and job requirements

Monitoring & Observability

You Can't Fix What You Can't See

Monitoring and observability are critical pillars of production DevOps. In this module, you'll implement industry-leading tools that give you complete visibility into your infrastructure, applications, and deployments — catching issues before they become outages.



Nagios

Classic infrastructure monitoring — track server health, network services, and set up alerts for downtime.



Prometheus

Open-source metrics collection and alerting toolkit, the backbone of modern cloud-native observability.



Datadog

Enterprise-grade full-stack monitoring with APM, log management, and distributed tracing capabilities.



CloudWatch

Native AWS monitoring for EC2, RDS, Lambda, and custom application metrics with alarm configuration.



Grafana

Build stunning real-time dashboards that visualize metrics from Prometheus, CloudWatch, and beyond.



Jira & ServiceNow

Understand ITSM workflows, incident management, and agile project tracking tools used in enterprises.

GitHub Actions — Modern CI/CD

GitHub Actions brings CI/CD directly into your repository. Build, test, and deploy your code automatically with every push — no external CI server required. It's the fastest-growing automation platform in modern DevOps.

Foundations

- Workflow structure & YAML syntax
- GitHub Marketplace actions
- Hosted & self-hosted runners
- Environment variables & secrets

Advanced Automation

- Reusable workflows
- Docker build & push automation
- Terraform integration
- Security & compliance checks

Kubernetes Deployments

- K8s & Helm deployments via Actions
- Real-time deployment scenarios
- Deployment strategies (blue/green, rolling)
- Notifications & integrations

AIOps — Intelligent Operations

DevOps got the pipeline shipping reliably — it didn't solve what happens at 3 a.m. when something breaks. AIOps adds a reasoning layer on top of the pipeline you already built, moving incident response along a maturity ladder from reactive to predictive to autonomous.

→ **Reactive**

Classic DevOps — an alert fires, a human investigates and fixes it.

→ **Predictive**

AIOps catches the anomaly before it crosses the alarm threshold.

→ **Autonomous**

Known-pattern incidents get fixed without paging anyone.

1

Foundations & the AWS AI Stack

Amazon Bedrock and MCP for live infrastructure access, plus a RAG-grounded runbook assistant that stops the model from hallucinating on ops questions.

2

The Intelligence Layer

Log-to-RCA generation that scores model output as a hypothesis, and an autonomous triage agent with a human approval gate on anything severe.

3

Automation & Reliability

Self-healing remediation bots with a full audit trail, plus evaluating agent output with precision, recall, F1, cost, and latency.

4

Production Hardening & Capstone

Budget alarms, least-privilege IAM, and shadow-mode rollouts — then a live capstone: Detect → Ground → Decide → Guard → Act.

20**Sessions**

One hour each, across 4 weeks

7**Hands-On Projects**

From an MCP assistant to a live capstone

100%**Real AWS Infra**

Every project ships against a live pipeline

Why This Program Stands Out

This isn't just another video course. The AWS DevOps Master Program is a live, interactive, job-focused training experience designed to take you from beginner to industry-ready professional in 4–6 months.

6

Months Duration

4–6 months of intensive live interactive training

6

Days/Week

Daily classes Mon–Fri with doubt sessions Mon–Sat

24/7

Support

Round-the-clock support so you're never stuck alone



Live Interactive Classes

Real-time learning with an expert instructor — ask questions, get answers instantly.



Weekly Assessments

Regular quizzes and assessments ensure you're retaining knowledge and staying on track.



Mock Interviews

Practice real interview scenarios with expert guidance to maximize your hiring chances.



Real-Time Projects

Build a portfolio of production-grade projects that prove your skills to employers.

Everything You Need to Land the Job

We don't just teach tools — we build careers. Every element of this program is designed to maximize your employability and ensure you walk into interviews with confidence.



Flexible Scheduling

Daily classes at 7:00 PM – 8:30 PM (IST). Q&A sessions at 10 AM and 4 PM. Attend live or catch recordings — the choice is yours.



2-Year Recording Access

Every session is recorded. Access 2 years of class recordings, plus batch transfer and repetition options if life gets in the way.



Career Support

Get expert resume building assistance, LinkedIn and Naukri profile optimization, and interview preparation materials curated for top DevOps roles.



Notes & Study Materials

Comprehensive notes, cheat sheets, and curated resources provided for every module — so you focus on learning, not note-taking.



Your DevOps career starts here. Join hundreds of students who have already transformed their careers with the AWS DevOps Master Program. Limited seats available — enroll today and take the first step toward a high-paying DevOps role.

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